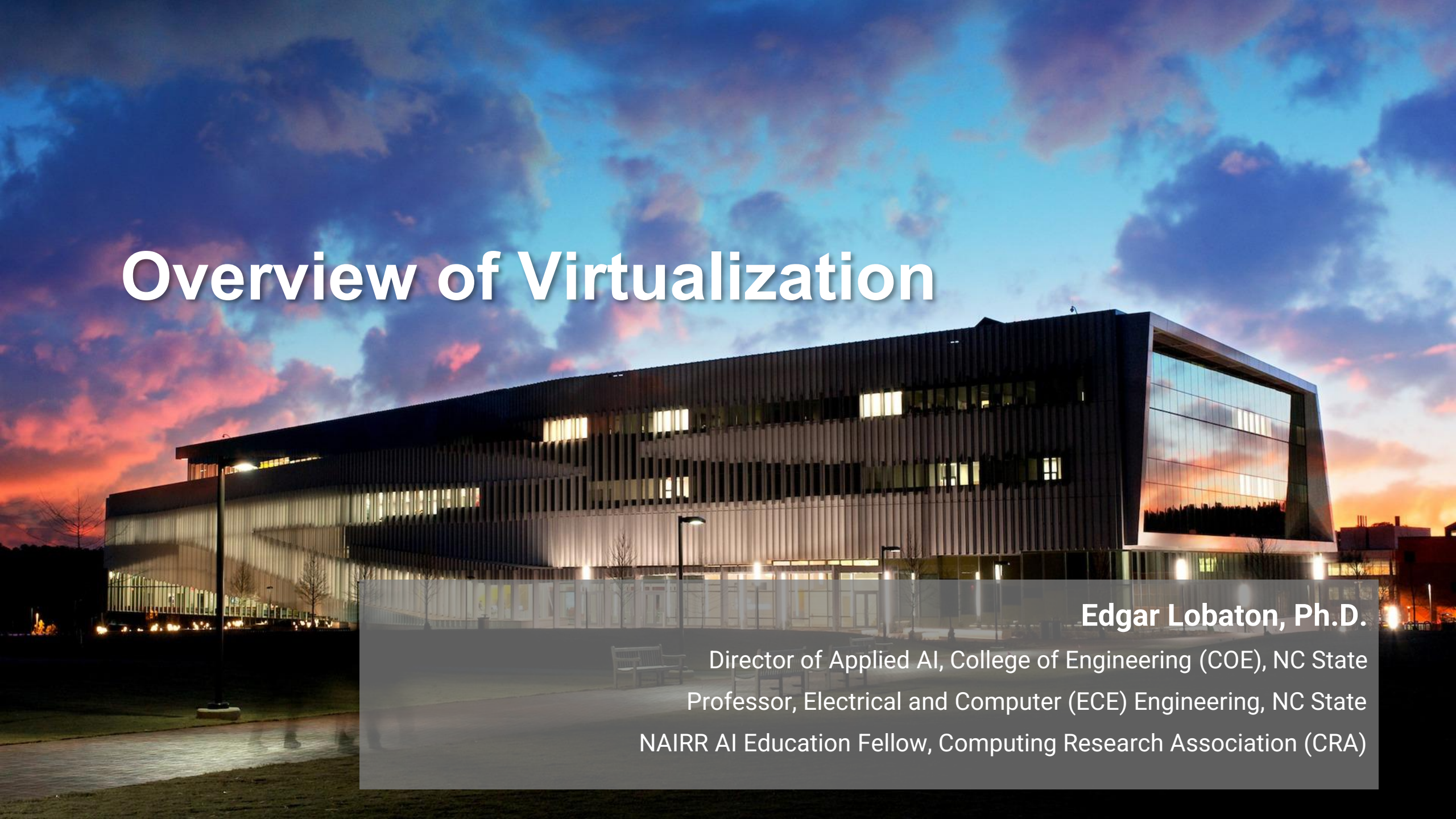
Overview of Virtualization



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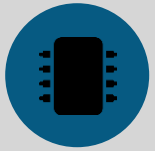
Why Virtualization Matters

Conflicting Package Versions

Two projects need different versions of the same library on one machine.



CUDA / Driver Mismatches

An installed PyTorch build doesn't match the local CUDA version.

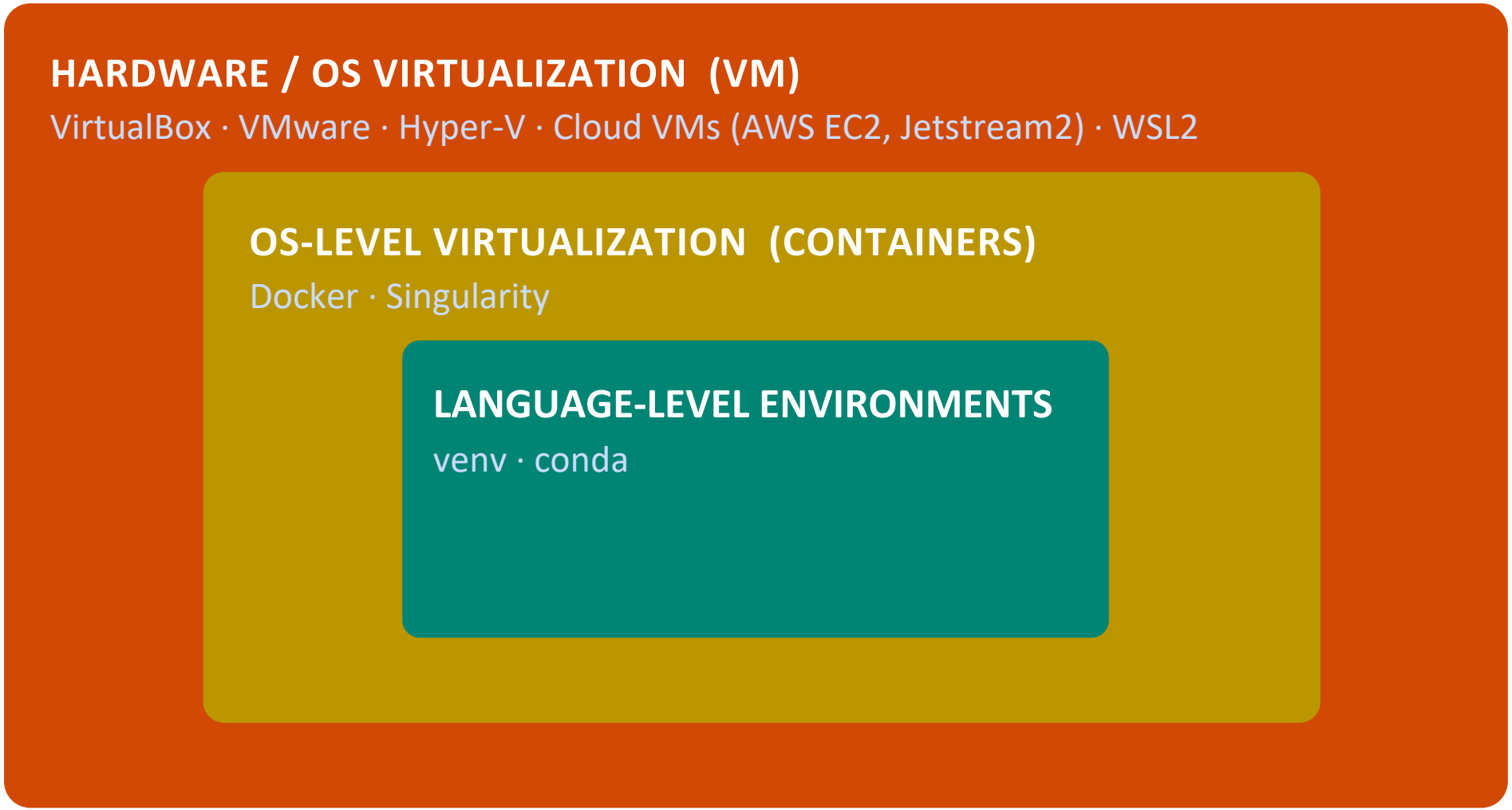
“Works on My Machine”

A script runs fine locally but breaks on a colleague's setup.

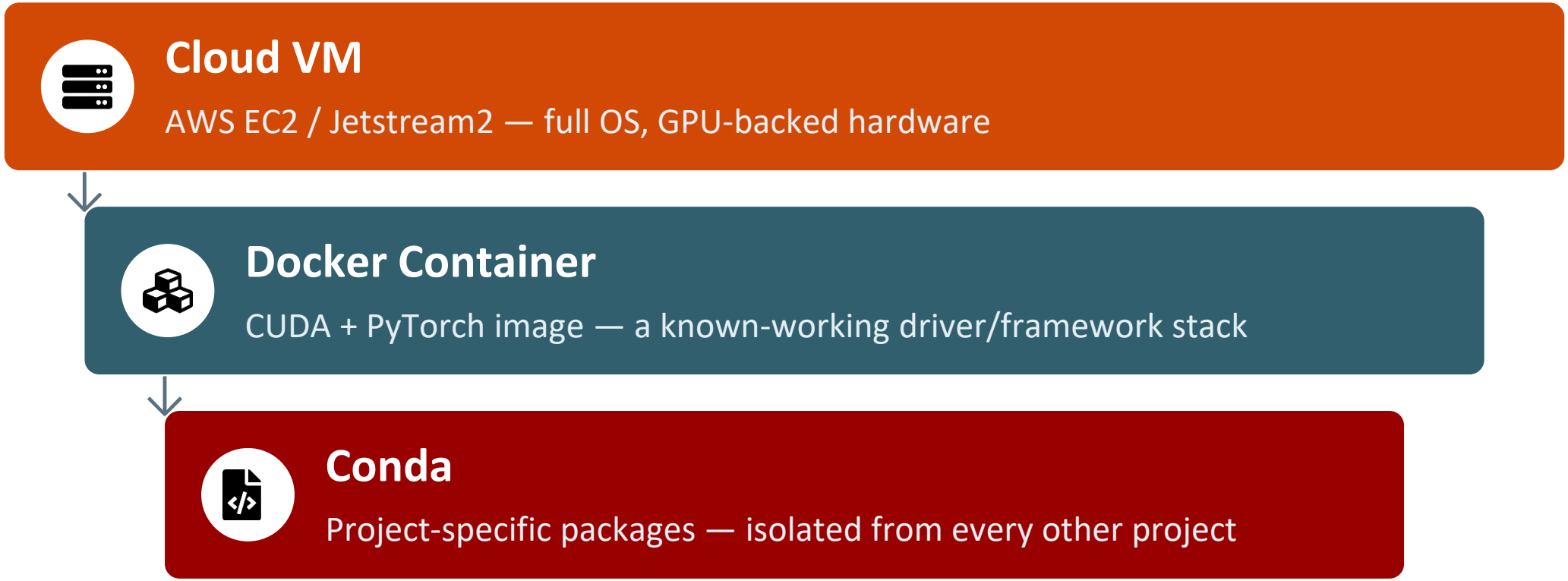
Shared Lab Hardware

Multiple users on one GPU workstation or Pi cluster collide.

Three Layers of Isolation



The Stack in Practice



These layers aren't competing choices — knowing which one is failing is the real skill.

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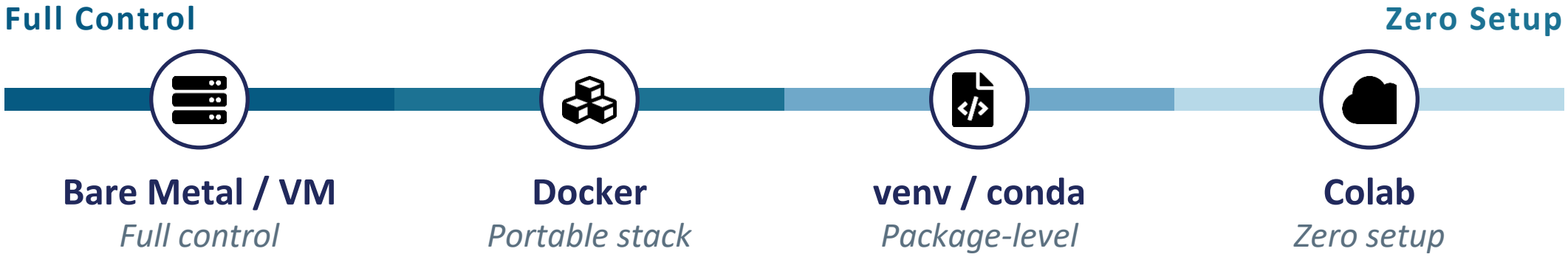
Where Does Google Colab Fit?



Colab is someone else's VM, pre-configured for you. Zero setup and tied to a Google account participants already have — ideal for workshops and demos, at the cost of less control over the underlying environment.

Which Layer Do You Need?

Just managing Python package versions?	**venv / conda**
Sharing a full stack with collaborators?	**Docker**
Deploying to HPC or a shared cluster?	**Singularity / Apptainer**
Need full hardware isolation or GPU cloud access?	**VM (local or cloud)**
Want zero setup for a workshop or demo?	**Colab**

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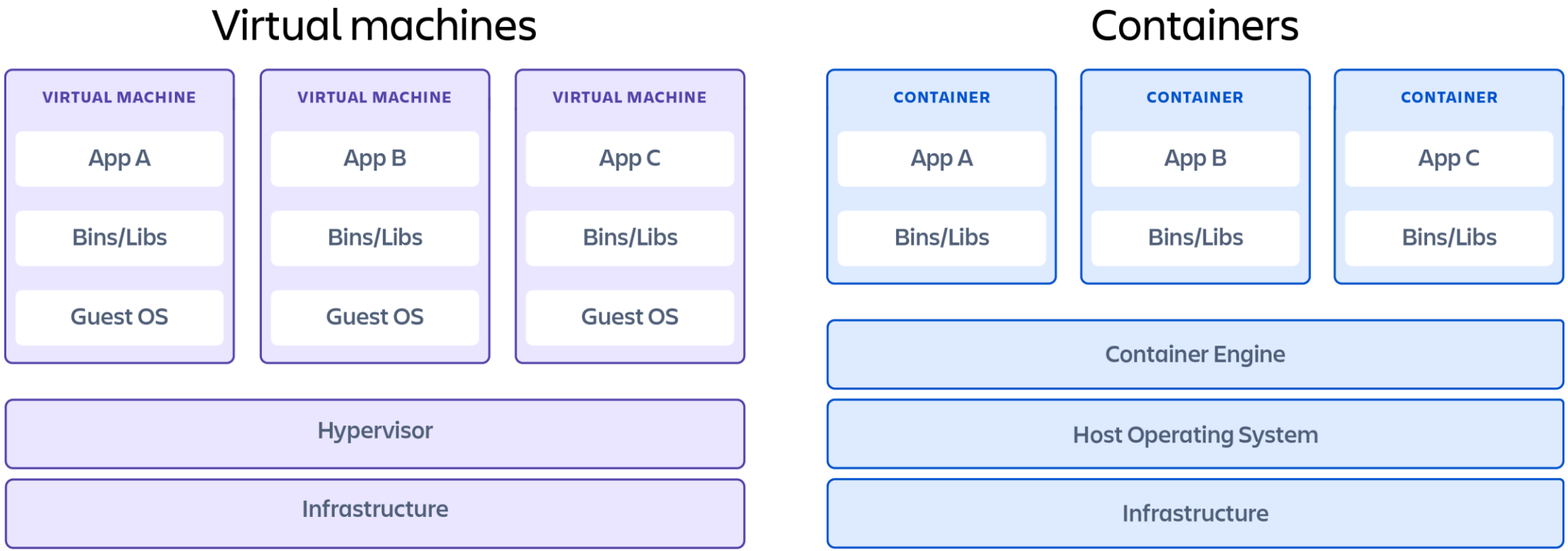
Hardware and OS Level Virtualization

VMs and Containers

Virtual Machines vs. Containers



Orchestrates the interactions between the hardware and the VMs

* https://www.atlassian.com/microservices/cloud-computing/containers-vs-vms

Virtual Machines

VMs virtualize an entire machine down to the hardware layers including CPU, Disk, Memory, and Network devices.

Pros:

- Full isolation security with vulnerabilities with the single system
- Beyond the basic hardware definition, the VM can be treated as a barebone computer. Good depending on your project's hardware req.

Cons:

- Time consuming to build and regenerate (full stack system)
- More storage, resources and booting time needed

Containers

Containers only virtualize software layers above the OS layer

Pros:

- Lightweight and fast to modify and iterate. Fast to load as well.
- Robust ecosystem with software applications such as databases, messaging systems and scientific software. Useful for microservices.

Cons:

- Vulnerabilities of a container can lead to system vulnerabilities.

Recommendations

- For scientific computing, containers are recommended since they are lightweight and ideal for reproducibility (e.g., packing a predictive AI model, or a visual 3D reconstruction model)

- The **Virtual Computing Lab (VCL)** relies on virtual machines
- The **HPC at NC State** will rely on containers

- For cloud deployments you may need a hybrid approach
- **Kubernetes** is used for scaling deployment of containerized apps

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Software-Level Virtualization

venv and conda

Dependency Hell

- Different projects may require different versions of the same library
- This may lead to broken code and difficulty sharing it

PIP – Pip Installs Packages

- It is the standard package manager for Python
- It makes it easy to install, upgrade and uninstall packages from the **Python Package Index** (PyPI), a Python software repository
- It automatically handles dependencies
- Works with *requirements.txt* allowing to replicate environments

Environments to the Rescue

- Different projects may have different libraries and dependencies… **Environments** are the solution.
- You can use the *requirements.txt* file to export the packages used and reproduce in a new environment.
- This helps keep your Python installation clean

Always use environments for every new project

Environments to the Rescue

VENV – Virtual Environments

- Built-in tool for creating environments.
- It creates a dedicated directory containing a copy of the Python interpreter and a pip installer.
- Any packages installed within this environment will be installed in the dedicated directory, leaving the global Python clean.
- Create: `python -m venv <directory_name>`
- Activate: `source <directory_name>/bin/activate`

Environments to the Rescue

CONDA – Package and Environment Manager

- It is like pip but it can handle Python, R, Java, C++, etc. environments
- It handles more complex dependencies using Conda-Forge repo
- Requires a separate installation: Anaconda or Miniconda

- Platforms such as **Anaconda** are great for scientific computing including environments with the basic data science libraries.
- Create: `conda create -n <env_name> python=<version>`
- Activate: `conda activate <env_name>`

Recommendations

- **Always use environments for every new project**
- Use conda first for package installation
- Use pip as a back up within a conda environment
- Conda is smart enough to track packages installed with pip as well

- Make sure to create *requirements.txt* files within repos for reproducibility.
- Conda has an equivalent *yml* file used to reproduce environments.

Let’s Recap